Smart Traffic Management System

Revolutionizing urban mobility through intelligent, sensor-driven traffic control.

The Problem

Traffic Congestion: A Global Challenge

Current urban traffic management largely relies on two outdated methods: **manual police intervention** and **fixed timer-based signals**. Both approaches are inherently inefficient and contribute significantly to widespread urban issues.

- **Manual Control:** Inconsistent, resource-intensive, and prone to human error.
- **Fixed Timers:** Unable to adapt to real-time traffic flow, leading to unnecessary delays and inefficient light cycles.

These inefficiencies manifest in critical problems:

- **Severe Congestion:** Increased travel times and productivity loss.
- **Wasted Resources:** Excessive fuel consumption and increased vehicle wear.
- **Environmental Pollution:** Higher emissions from idling vehicles.
- **Economic Impact:** Billions lost annually due to traffic delays.

We need an adaptive, sensor-based automated control system to alleviate these pressing issues.

Our Solution

Dynamic Traffic Optimization

Our objective is to design and implement a smart traffic management system that addresses the core limitations of existing solutions. This system will leverage advanced sensing and embedded processing to create a truly responsive and efficient traffic network.

Real-time Vehicle Detection

Utilize **ultrasonic sensors** to precisely detect vehicle presence and density in each lane, providing accurate, instantaneous data.

Dynamic Traffic Light Control

Algorithmically adjust traffic light timings based on live sensor data, optimizing flow and minimizing idle times.

Robust Embedded Control

Employ the **STM32F405RGT6 microcontroller** as the central processing unit, ensuring reliable and high-performance system operation.

Background & Innovation

Beyond Conventional Approaches

Our research involved an in-depth analysis of existing smart traffic solutions and academic literature. While the field is evolving, a critical gap remains that our system aims to address.

Current Trends in Smart Traffic

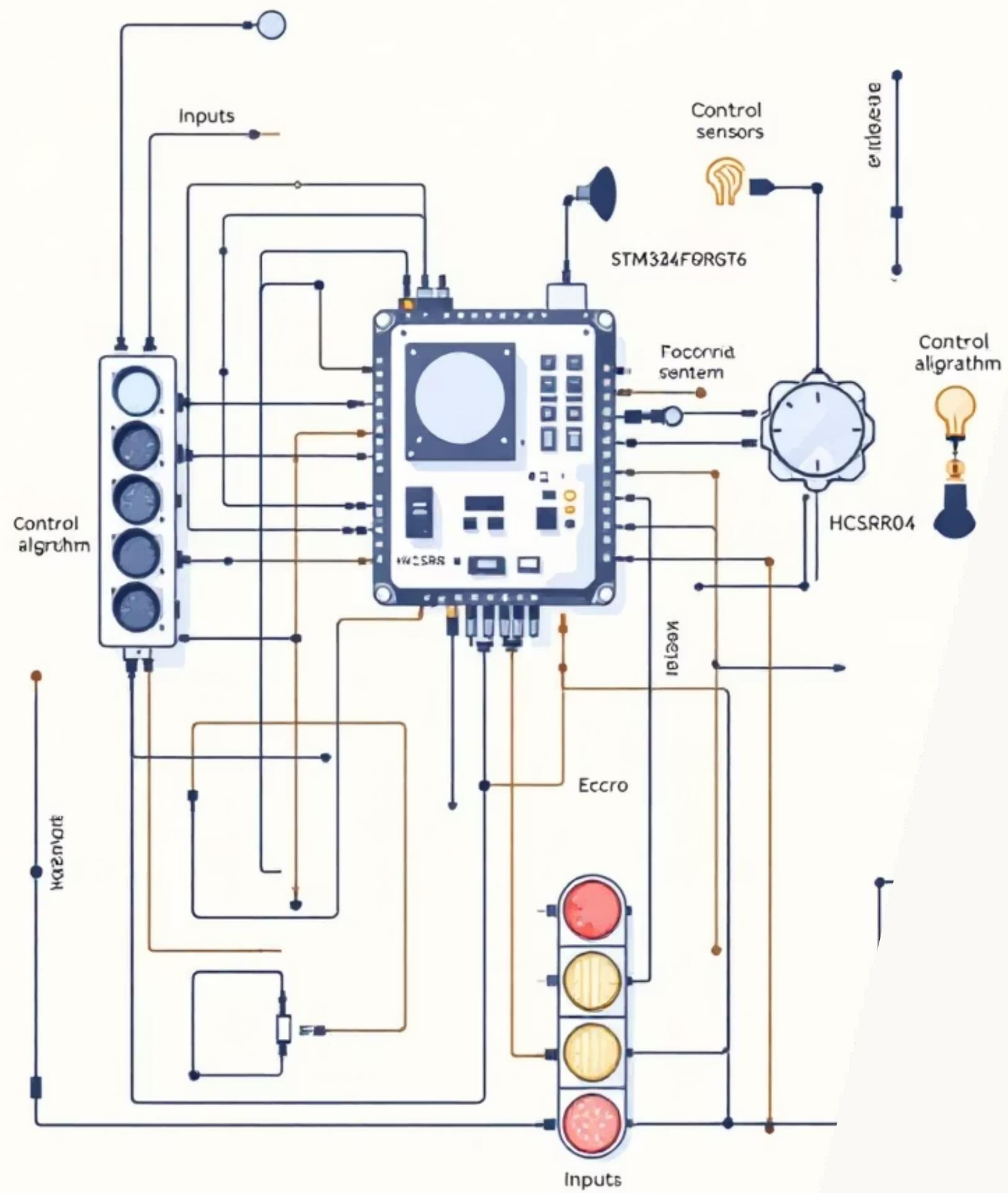
- IoT-based Traffic Systems (*Ref: IEEE Xplore, 2023*)
- AI-driven Adaptive Traffic Management (*Springer, 2022*)
- Sensor-based Signal Manipulation (*MDPI, 2024*)

Many cutting-edge solutions predominantly rely on **camera and AI technologies** for vehicle detection and analysis.

The Cost Barrier

While effective, camera and AI-based systems often come with a **prohibitive cost** due to high-resolution cameras, powerful processing units, and complex AI model training. This limits their widespread adoption, especially in resource-constrained environments.

Our proposed system provides a **low-cost ultrasonic sensor** alternative, offering a scalable and economically viable solution without compromising on adaptive control capabilities.



System Architecture Overview

Integrated Hardware & Software

Inputs

Ultrasonic Sensors (**HCSR04**) strategically placed on each lane for accurate presence detection.

Processor

STM32F405RGT6 microcontroller, the brain of the system, responsible for real-time data processing and decision-making.

Outputs

Standard **LED Traffic Lights** (Red, Yellow, Green) for visual signaling to drivers.

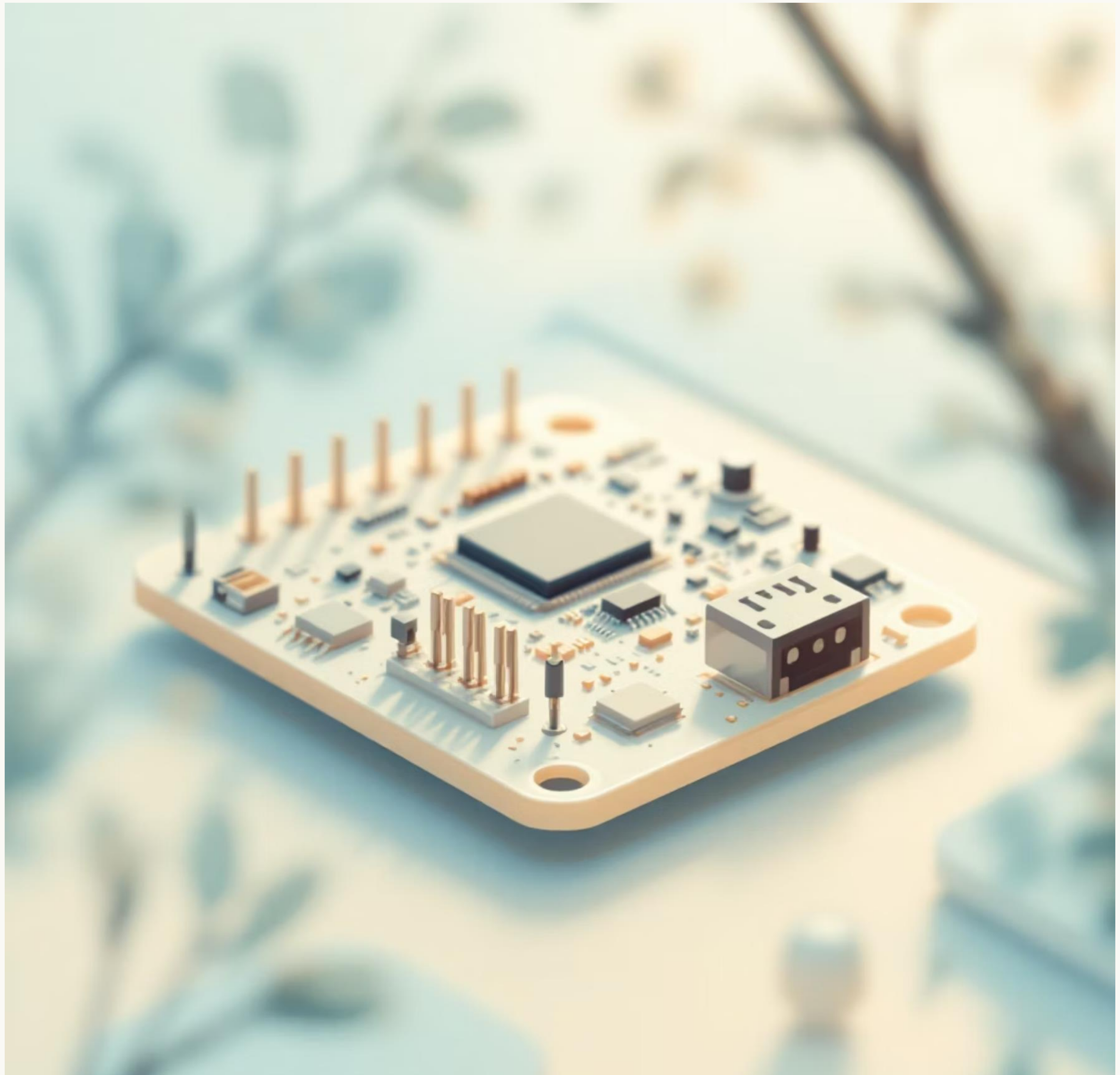
Control & Timing

Custom **Dynamic Priority Assignment** algorithm and **RCC/TIM2** for precise timing and reliable operation.

Core Hardware Components

Robust and Efficient Selection

The selection of hardware components is critical for the system's performance, reliability, and cost-effectiveness. We've chosen industry- and education-standard components known for their capabilities and ease of integration.



Software Development

Baremetal Efficiency & Modularity

Our software is engineered for maximum efficiency and control, developed using baremetal C code for direct hardware interaction. This approach ensures minimal overhead and optimal performance, critical for real-time applications.

Baremetal C Code

Direct register access for low-level control, eliminating OS overhead.

Key Driver Development

- **GPIO:** Configures pins for input (sensors) and output (LEDs).
- **Timer (TIM2):** Provides precise delays and measures ultrasonic echo duration.
- **Ultrasonic:** Handles trigger pulse generation and distance calculation.

Traffic Light State Machine

Manages the sequence and timing of traffic light changes based on the control algorithm.

Agile Lifecycle

Iterative development for rapid prototyping, testing, and continuous improvement.

Flow of Control

Real-time Decision Making Process

The system operates on a continuous feedback loop, adapting to traffic conditions in real-time. This sequential flow ensures dynamic and efficient traffic management.

01

Trigger Ultrasonic Sensors

The microcontroller initiates a trigger pulse from each sensor to emit ultrasound waves.

02

Measure Echo Pulse

The time taken for the ultrasound wave to return (echo) is precisely measured by the microcontroller's timer.

03

Calculate & Compare Distance

The echo duration is converted into distance. This distance is then compared against a predefined threshold to determine vehicle presence.

04

Assign Traffic Light Priority

Based on vehicle detection across all lanes, the system's algorithm dynamically assigns priority to the busiest lane(s).

05

Update Light States

Traffic lights are updated according to the assigned priorities, optimizing flow and reducing wait times.

Code Logic Summary

Algorithmic Core of STM32

Initialization Sequence

- **RCC Clock Enabled:** System clock configured for optimal performance.
- **GPIOs & TIM2 Configured:** All necessary pins and timer peripherals initialized.
- **Default State:** All traffic lights are initially set to red, ensuring a safe startup.

Ultrasonic Measurement Logic

- **Trigger Pulse:** A short pulse is sent to the sensor's trigger pin.
- **Echo Duration:** The microcontroller listens for the echo pulse on the sensor's echo pin, measuring its duration using TIM2.
- **Distance Calculation:** Distance is computed using the formula: `Distance = Pulse width / 58` (for cm).

Traffic Control Algorithm

- **Vehicle Detection:** If the calculated distance is less than a predefined threshold, vehicles are considered present in that lane.
- **Dynamic Priority:** Lanes with detected vehicles are given priority, overriding the default cyclic sequence.
- **Green Time Management:** A prioritized lane receives a maximum green time before transitioning to yellow and then red.
- **Next Lane Selection:** After a lane's cycle, the algorithm evaluates other lanes for priority or proceeds in a default order if no priority is needed.

Continuous Operation Loop

- Sensors continuously update distance measurements.
- The traffic control algorithm re-evaluates and updates the light cycle dynamically based on the latest data.

Prototype Implementation

Bringing the Concept to Life

Our functional prototype demonstrates the viability and effectiveness of the Smart Traffic Management System. This hands-on implementation phase allowed us to rigorously test and validate our design.

Physical Layout	Breadboard connections facilitating easy assembly and modification.
Microcontroller Integration	STM32F405RGT6 seamlessly connected to all sensors and output LEDs.
Sensor Configuration	Four ultrasonic sensors strategically positioned to simulate traffic lanes.
Traffic Signal Representation	LEDs clearly represent red, yellow, and green traffic lights.
Power Supply	Reliable power provided via USB or a regulated external source.
Testing Environment	Real-time testing conducted using toy vehicles to simulate varying traffic conditions and observe dynamic responses.

Code Implementation

Core Logic Snippets

GPIO & TIM2 Setup

```
// GPIO for HC-SR04 Trigger & EchoHAL_GPIO_Init(...);// TIM2 for
microsecond delaysTIM_HandleTypeDef htim2;htim2.Instance =
TIM2;HAL_TIM_Base_Init(&htim2);HAL_TIM_Base_Start(&htim2);
```

Ultrasonic Measurement

```
float getDistance_cm() {    // Trigger pulse
HAL_GPIO_WritePin(TRIG_GPIO_Port, TRIG_Pin, GPIO_PIN_SET);
delay_us(10);    HAL_GPIO_WritePin(TRIG_GPIO_Port, TRIG_Pin,
GPIO_PIN_RESET);    // Measure echo pulse duration
while(HAL_GPIO_ReadPin(ECHO_GPIO_Port, ECHO_Pin) == GPIO_PIN_RESET);
uint32_t start_time = __HAL_TIM_GET_COUNTER(&htim2);
while(HAL_GPIO_ReadPin(ECHO_GPIO_Port, ECHO_Pin) == GPIO_PIN_SET);
uint32_t end_time = __HAL_TIM_GET_COUNTER(&htim2);    return (end_time -
start_time) * 0.034 / 2; // Speed of sound in cm/µs}
```

Traffic Cycle Logic

The system prioritizes traffic flow based on real-time vehicle detection. When a vehicle is detected, the corresponding lane's green light duration is extended. In the absence of traffic, a default cycle maintains flow, preventing unnecessary delays.

Project Management

Agile Development Lifecycle

01

Requirement Gathering

Defined core functionalities: real-time detection, adaptive timing, and basic traffic states.

02

Sprint Planning

Structured development into two-week sprints, focusing on hardware integration, sensor logic, and traffic control algorithms.

03

Incremental Prototyping

Developed modular code for GPIO, timers, and sensor readings, integrating and testing in stages.

04

Testing & Feedback

Conducted bench testing and simulated traffic scenarios to refine sensor accuracy and timing logic.

05

Deployment

Finalized prototype for demonstration, showcasing adaptive behavior in varied traffic conditions.

Our agile approach facilitated rapid iteration and allowed us to adapt to unforeseen technical challenges efficiently, leading to a robust and functional prototype.

Prototype Evaluation

Key Advantages & Benefits

1

Reduced Congestion

Dynamic adjustment of signal timings based on real-time traffic flow minimizes unnecessary waits.

2

Fuel & Time Savings

Optimized traffic flow leads to less idling, conserving fuel and reducing commute times for drivers.

3

Pollution Control

Less idling directly translates to lower vehicular emissions and improved urban air quality.

4

Low-Cost Implementation

Utilizes readily available and economical components like STM32 microcontrollers and ultrasonic sensors.

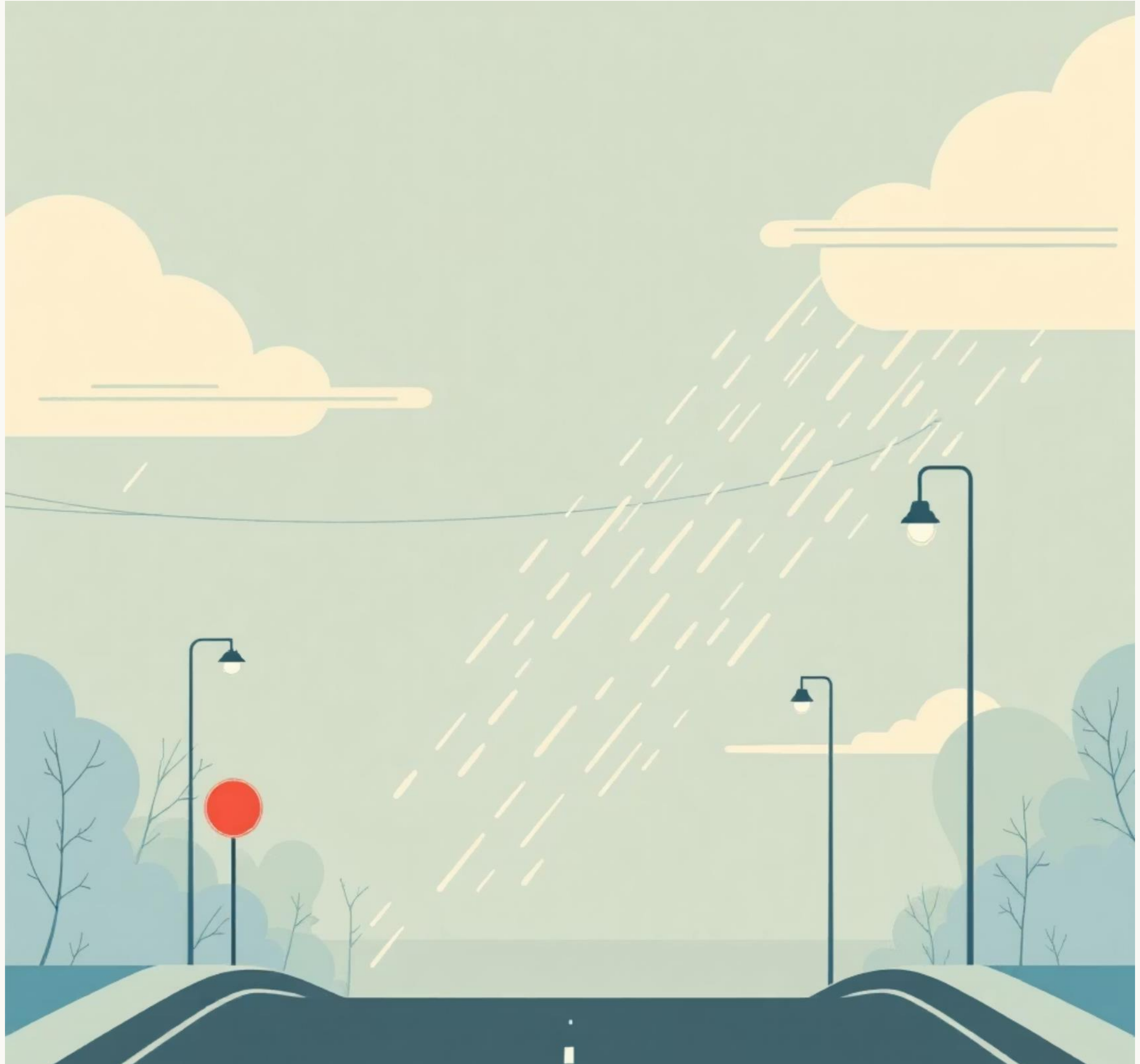
5

Scalable

The modular design allows for integration into larger smart city infrastructure, potentially with AI/IoT capabilities.

Prototype Evaluation

Challenges & Limitations



- Sensor accuracy can be **affected by adverse weather** conditions such as heavy rain or dense fog, leading to inconsistent readings.
- The HC-SR04 ultrasonic sensor has a **limited effective detection range** of approximately 400 cm, restricting its application in wide intersections.
- Scaling the system to manage traffic in large, complex urban environments would necessitate a robust **IoT backbone** for centralized data processing and control.
- The current prototype **requires continuous power** and would need backup power solutions (e.g., UPS) to maintain operation during electrical blackouts.

Future Outlook

Beyond the Prototype

IoT Cloud Integration

Connecting units to cloud platforms for centralized data analytics, remote monitoring, and firmware updates.

AI-Based Prediction

Implementing machine learning algorithms to predict traffic density based on historical data and real-time inputs.

Solar Power

Integrating solar panels and battery storage to create self-sufficient, environmentally friendly traffic units.

Conclusion

The STM32-based adaptive traffic control system offers a feasible, cost-effective, and efficient solution for modern urban traffic management, reducing congestion and human intervention. Further enhancements can leverage IoT and AI for even smarter cities.

References: IEEE Xplore: Smart Traffic Systems (2023) | MDPI Sensors: Adaptive Signal Control (2024) | Springer: AI in Transportation (2022) | STM32F405RGT6 & HCSR04 Datasheets